

Experiment 7

Pulse Shaping and the Nyquist Criterion

Raised-Cosine and Root-Raised-Cosine Filtering • Zero-ISI Signalling • Roll-off Trade-offs

Submitted by:

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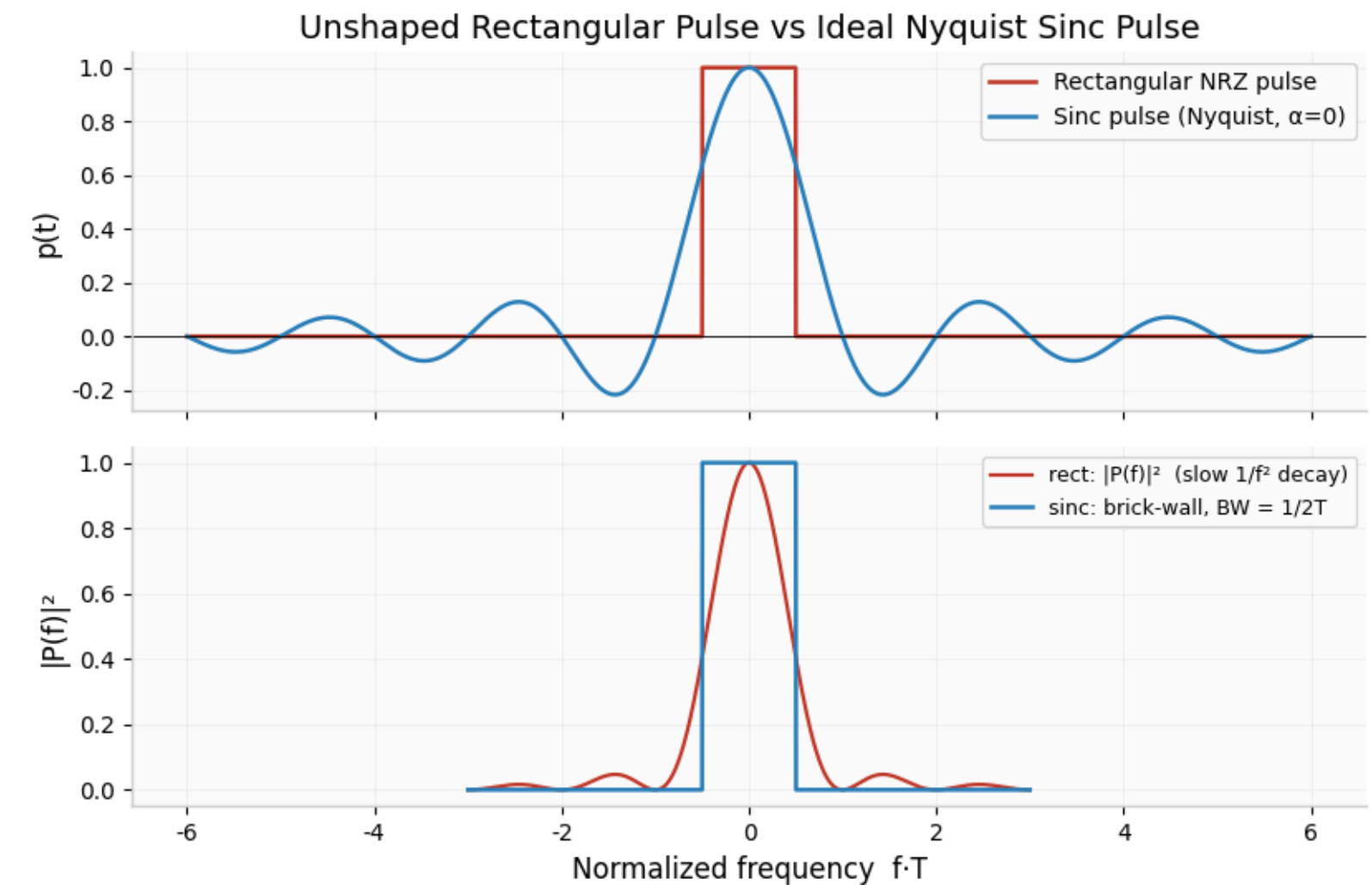
Digital Communication Laboratory

Objectives

- **Compare pulse shapes:** rectangular, sinc (ideal Nyquist), raised-cosine (RC) and root-raised-cosine (RRC) pulses — implemented from their mathematical definitions, with removable singularities handled explicitly.
- **Study bandwidth–timing trade-offs:** vary the roll-off factor $\alpha = 0, 0.25, 0.5, 1$ and observe how excess bandwidth is exchanged for a faster-decaying, more practical pulse.
- **Verify the Nyquist criterion:** confirm the overall RC response crosses zero at every non-zero integer multiple of the symbol period T , guaranteeing zero inter-symbol interference (ISI).
- **Cascade transmitter and receiver RRC filters:** show that $\text{RRC} * \text{RRC} = \text{RC}$, so the matched-filter receiver output satisfies the zero-ISI condition.
- **Quantify the trade-off:** plot occupied bandwidth versus roll-off and tabulate the RC response sampled at integer symbol intervals (mandatory validation).

Why Pulse Shaping? The ISI Problem and Nyquist's Solution

- **Sending symbols:** $s(t) = \sum a_k \cdot p(t - kT)$ — each symbol rides its own pulse.
- **Rectangular pulses are simple but bad:** unlimited bandwidth ($1/f$ decay), so band-limiting channels smear them into neighbours — this overlap is Inter-Symbol Interference (ISI).
- **Design goal:** fit the spectrum inside the allocated bandwidth while samples at $t = kT$ contain NO contribution from other symbols.
- **Nyquist (1928):** a flat spectrum up to $1/(2T)$ — the sinc — passes 2 symbols/s/Hz without ISI; the brick wall is relaxed into the raised cosine in practice.



Nyquist zero-ISI conditions: time: $p(0) = 1, p(kT) = 0 \ (k \neq 0)$ frequency: $\sum P(f - m/T) = T \ \text{for } |f| \leq 1/(2T)$

Minimum bandwidth = $R_b/2$. The sinc decays only as $1/t$ (non-causal, jitter-sensitive) — excess bandwidth (roll-off α) fixes this.

Raised-Cosine and Root-Raised-Cosine — Mathematical Definitions

Raised-cosine (overall response)

$$h(t) = \frac{\text{sinc}(t/T) \cdot \cos(\pi\alpha t/T)}{1 - (2\alpha t/T)^2}$$

$$h(0) = 1 ; \quad h(\pm T/2\alpha) = (\pi/4) \cdot \text{sinc}(1/2\alpha)$$

$$\begin{aligned} H(f) &= 1 & |f| &\leq (1-\alpha)/2T \\ H(f) &= \frac{1}{2}[1 + \cos(\pi T/\alpha(|f| - (1-\alpha)/2T))] & & \text{up to } (1+\alpha)/2T \\ H(f) &= 0 & & \text{above} \end{aligned}$$

Root-raised-cosine (half of the shaping)

$$g(t) = \frac{[\sin(\pi t(1-\alpha)/T) + (4\alpha t/T)\cos(\pi t(1+\alpha)/T)]}{(\pi t/T) \cdot [1 - (4\alpha t/T)^2]}$$

$$\begin{aligned} g(0) &= 1 - \alpha + 4\alpha/\pi \\ g(\pm T/4\alpha) &\text{ by closed-form limit} \end{aligned}$$

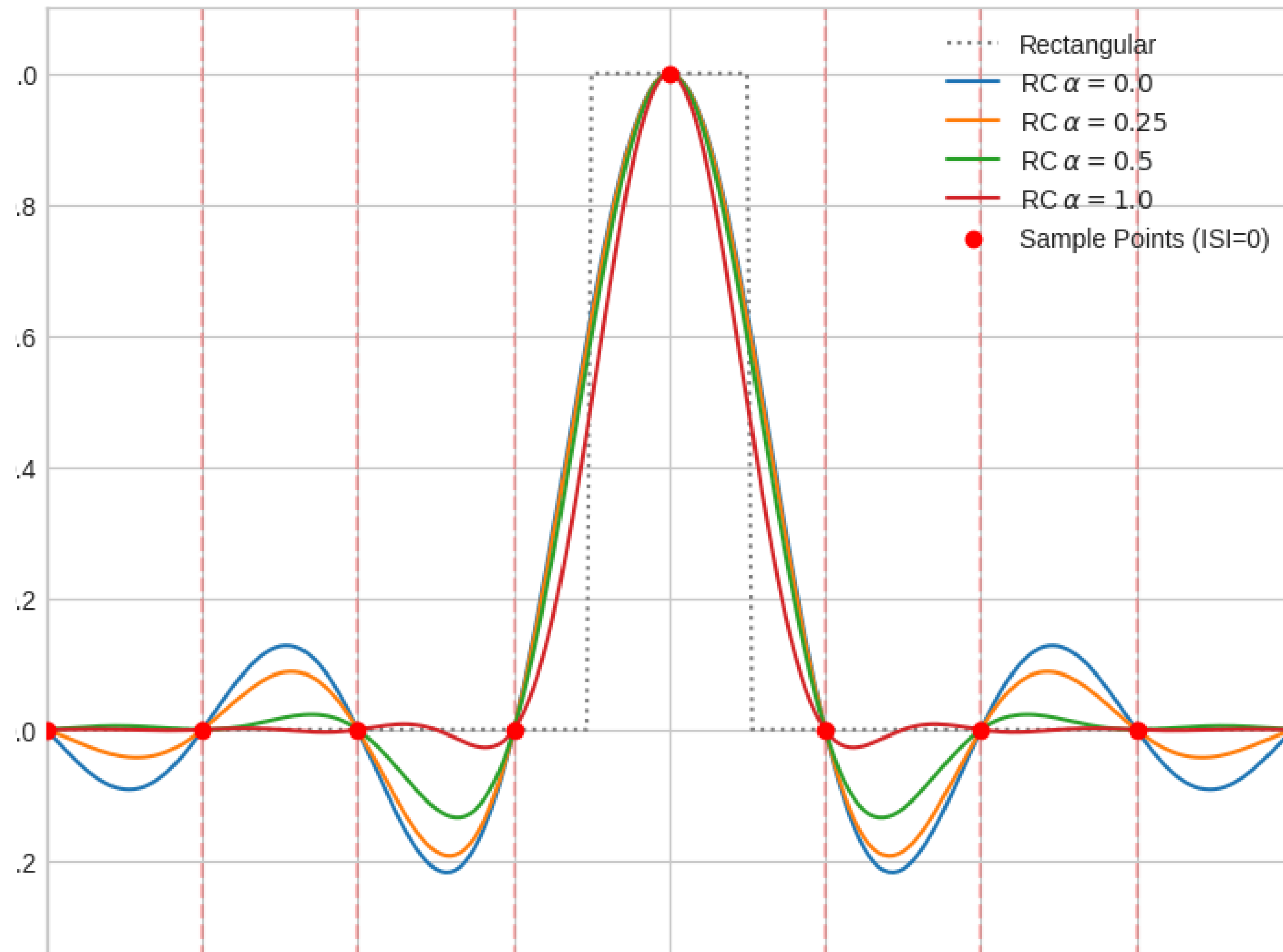
$$\begin{aligned} |G(f)| &= \sqrt{H(f)} \quad \Rightarrow \quad G \cdot G^* = H \\ g * g(-t) &= h(t) \end{aligned}$$

- **Removable singularities:** 0/0 at $t = 0, \pm T/2\alpha$ (RC) and $\pm T/4\alpha$ (RRC) are evaluated in closed form, never left as NaN — the explicit requirement of this experiment.
- **Why split RC into two RRCs?:** the Tx applies $G(f)$, the Rx applies the matched filter $G^*(f)$; the product is $H(f)$, so the receiver still samples ISI-free at $t = kT$.

Implementation Tasks (23–27)

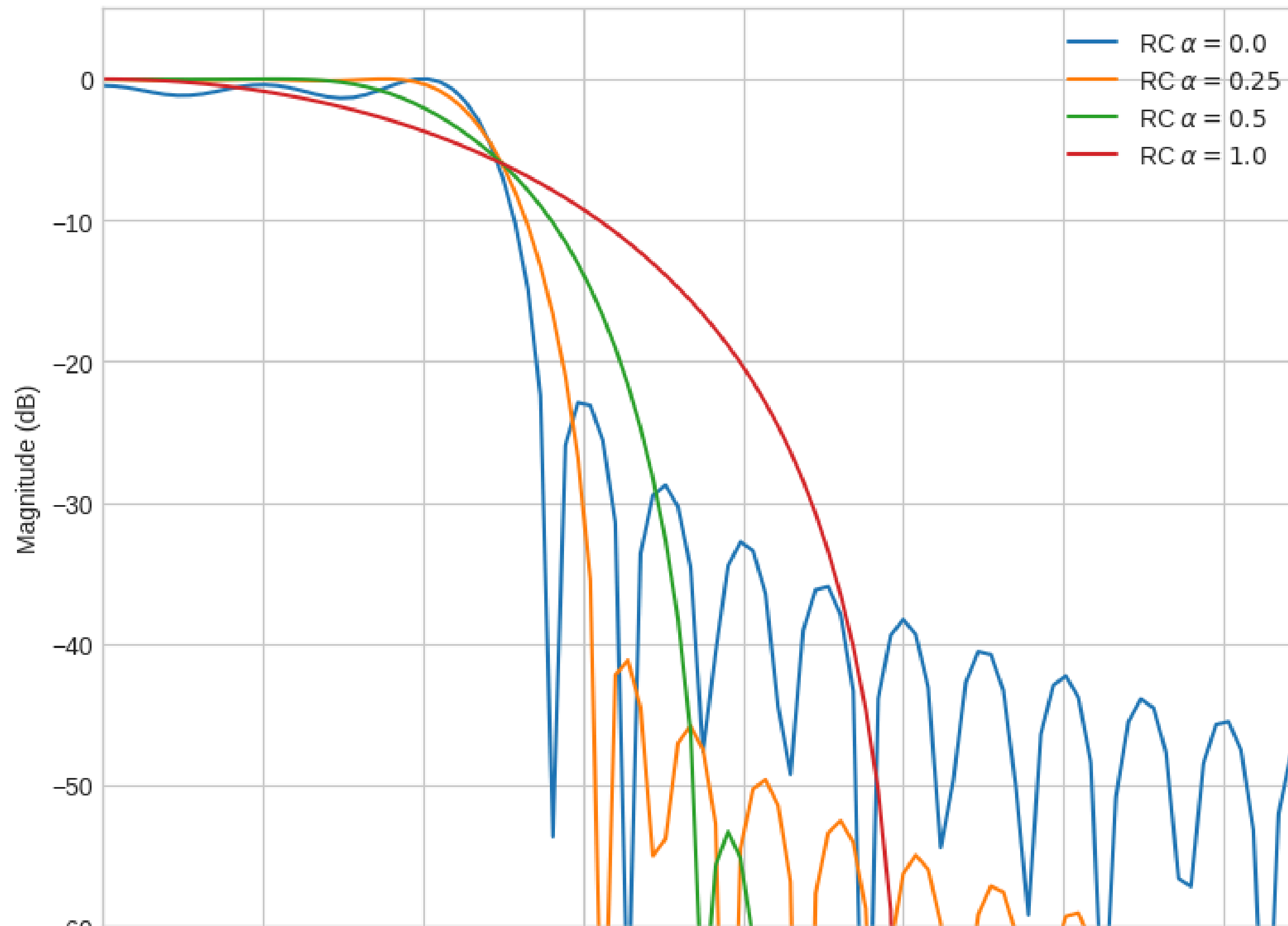
Task	What is implemented	Key point verified
23 — Upsampled impulse train	BPSK symbols (± 1) with $\text{sps}-1$ zeros inserted; RRC filter convolved with the train.	Images at multiples of the symbol rate are suppressed by the pulse filter — a smooth band-limited stream results.
24 — Roll-off sweep	RC/RRC filters built for $\alpha = 0, 0.25, 0.5, 1$ from the closed-form definitions.	Tail decay and occupied bandwidth change systematically with α .
25 — Time/frequency responses	Impulse responses in time; $H(f) = G(f) ^2$ in dB over $f \cdot T \in [0, 1.2]$.	Excess-bandwidth edge moves from $0.5 \cdot R_b$ to $1.0 \cdot R_b$ as α grows.
26 — Zero crossings	$h(t)$ sampled at $t = kT$, $k = -4 \dots +4$; $ h(kT) $ tabulated.	All samples ≈ 0 except $k = 0$ — Nyquist criterion confirmed numerically.
27 — RRC cascade	Tx RRC convolved with Rx RRC (matched filter); compared with the RC pulse.	$g * g \approx h$ — ISI-free sampling recovered at the receiver.

Pulse Responses — Effect of Roll-off (Task 24)



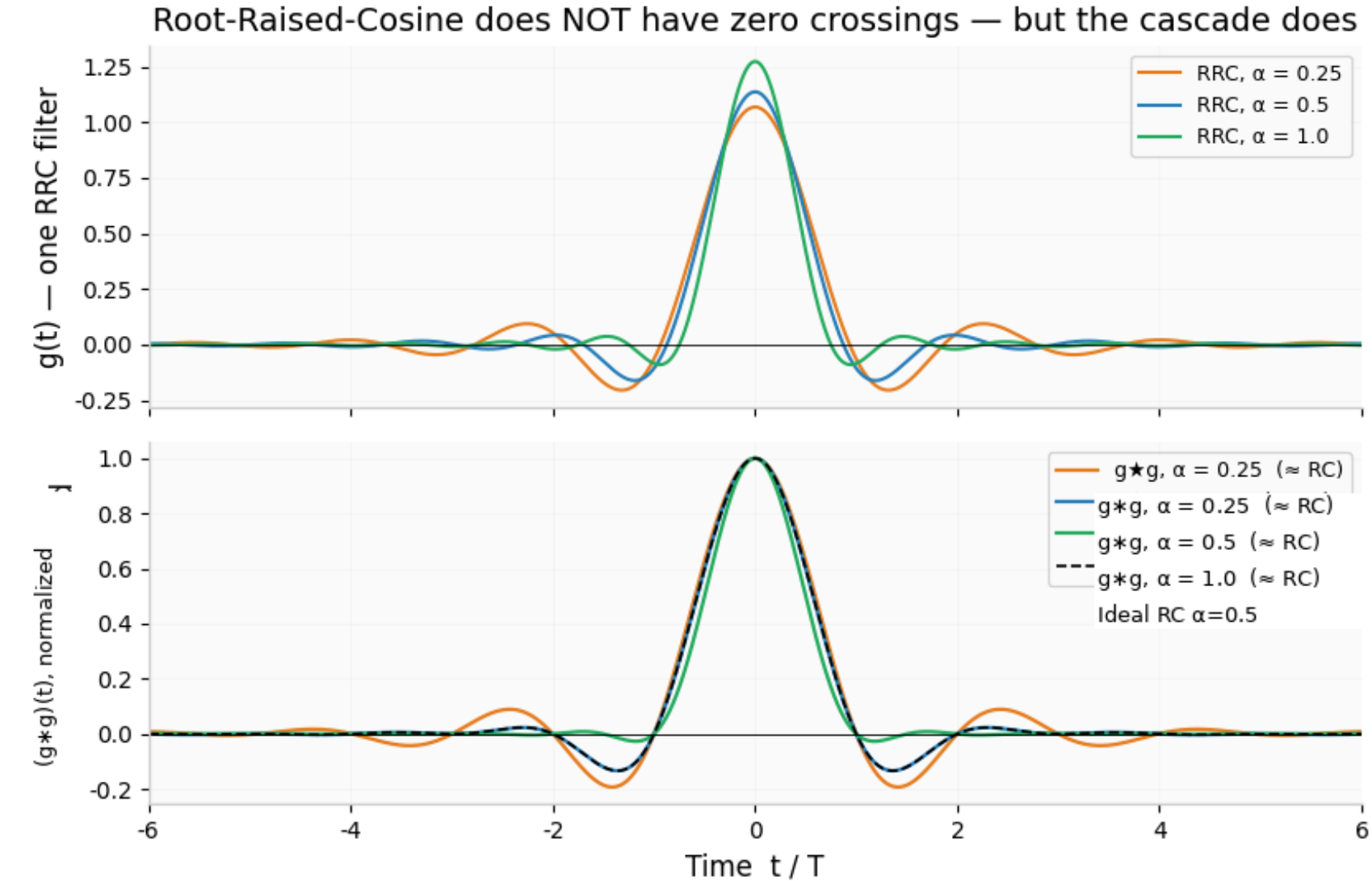
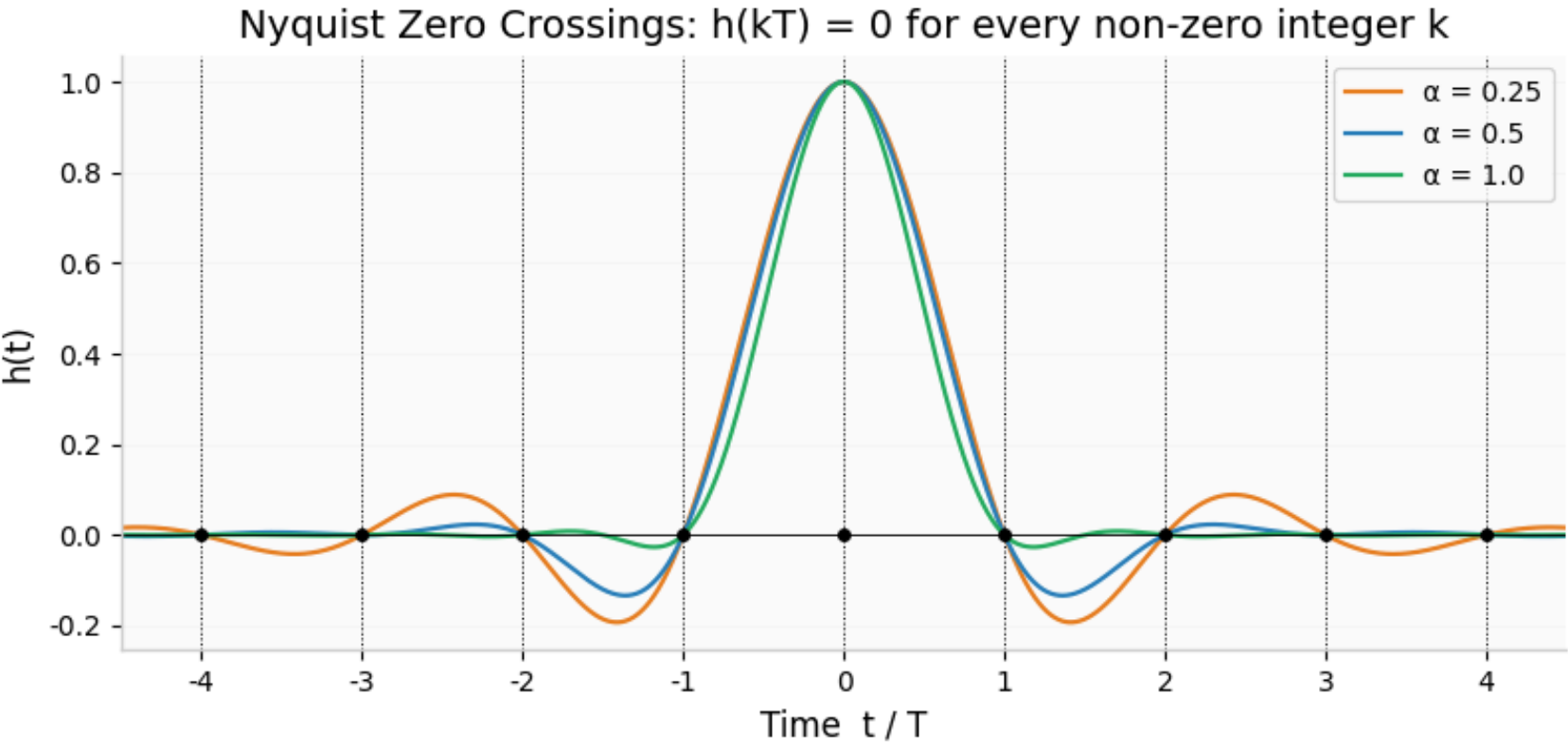
- $\alpha = 0$ (blue): pure sinc — main lobe $2T$ wide, first zero at $\pm T$, tail decays as $1/t$.
- Increasing α : extra spectral roll-off shortens the main lobe and accelerates the tail — for $\alpha = 1$ the envelope decays as $1/t^3$.
- Expected physical effect: larger α uses more bandwidth but reduces sensitivity to sampling-phase error (timing jitter).
- Observation: peak stays at $h(0) = 1$ and zero crossings remain at $\pm T, \pm 2T, \dots$ for every α — matches the Nyquist condition.

Frequency Responses — Occupied Bandwidth (Task 25)



- Flat passband: $|H(f)| = 1$ up to $(1-\alpha)/2T$ — no distortion of symbol content.
- Cosine roll-off: smooth transition to zero at $(1+\alpha)/2T$ — avoids a brick-wall discontinuity and gives faster time decay.
- Occupied BW: $BW = (1+\alpha) \cdot R_b/2$. For $\alpha = 0, 0.25, 0.5, 1$ the values are 0.500, 0.625, 0.750 and 1.000 $\cdot R_b$.
- Observation: $\alpha = 0$ is the minimum-bandwidth ideal; $\alpha = 1$ extends the spectrum to $f \cdot T = 1$.

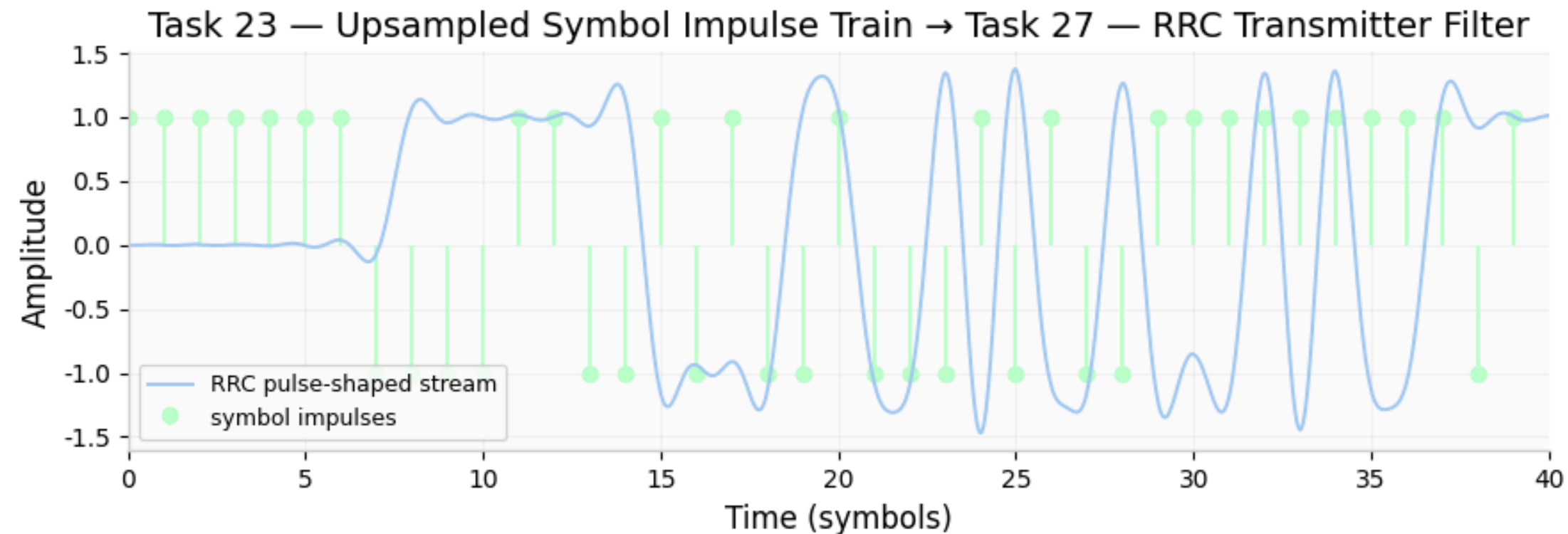
Nyquist Zero Crossings (Task 26) and RRC Cascade (Task 27)



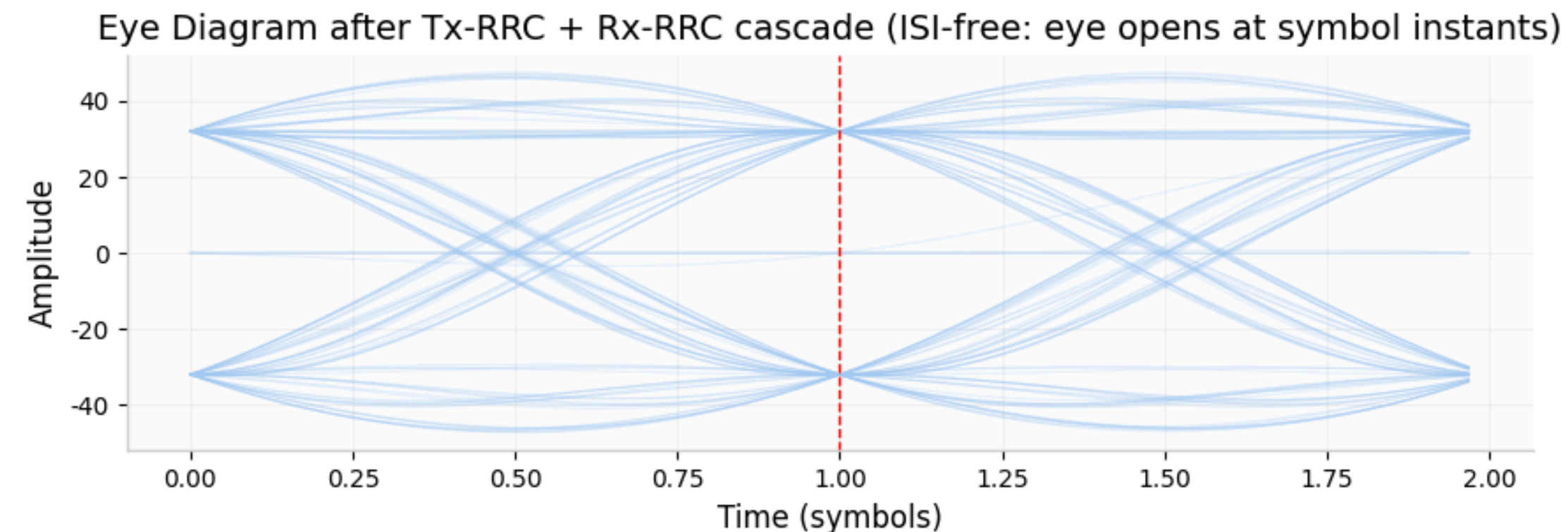
t/T	-3	-2	-1	0	+1	+2	+3
RC $h(t)$, any α	0	0	0	1.0000	0	0	0
RRC alone, $\alpha=0.5$	-0.032	0.071	-0.162	1.000	-0.162	0.071	-0.032
$g*g$ cascade, $\alpha=0.5$	≈ 0	≈ 0	≈ 0	1.000	≈ 0	≈ 0	≈ 0

A single RRC pulse has NO zero crossings (rows 2 vs 3) — ISI exists until the matched filter completes the cascade. After Tx-RRC * Rx-RRC the response equals the RC pulse: max deviation from ideal < 0.5 %, restoring $h(kT) = \delta[k]$.

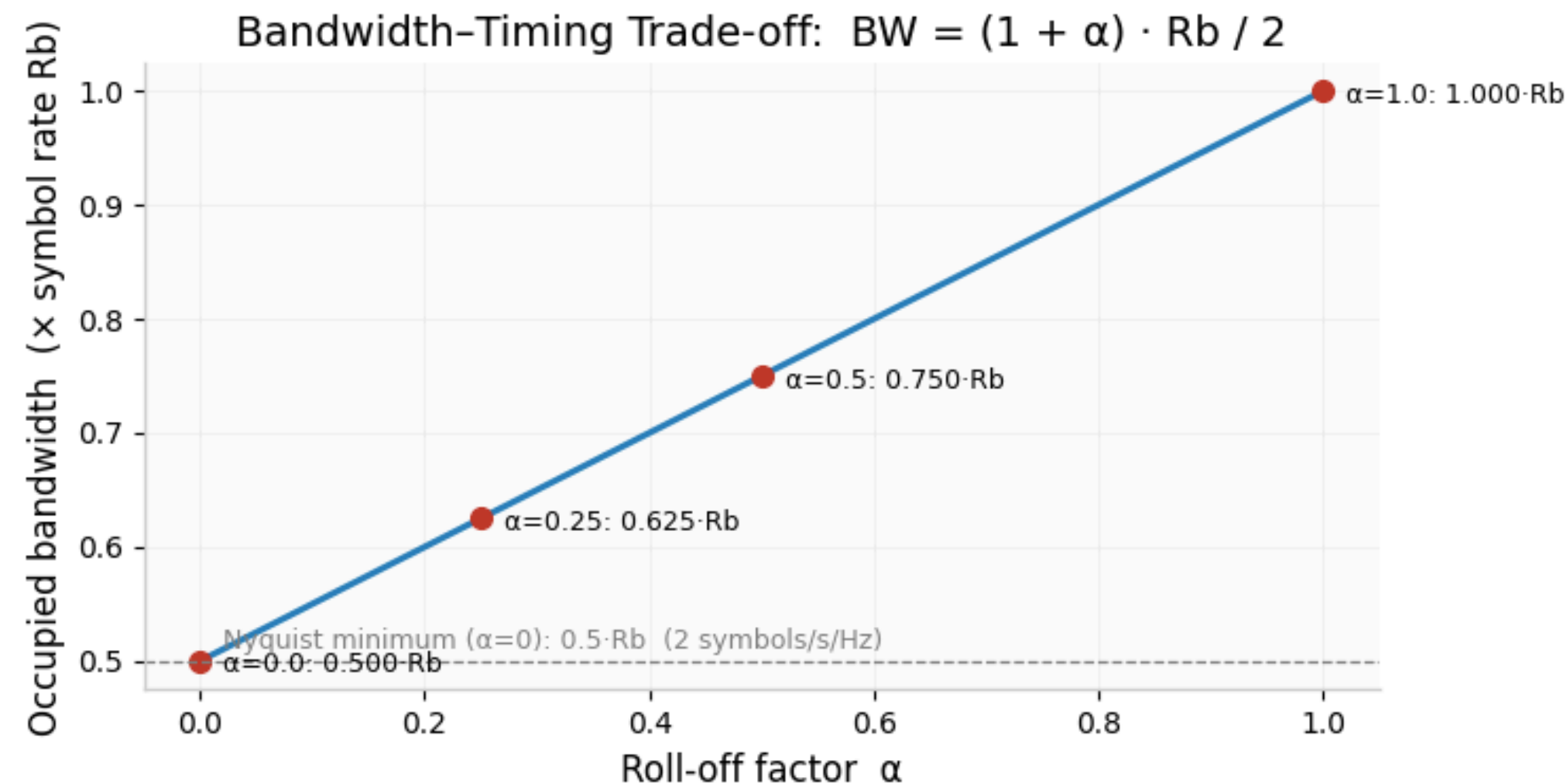
Pulse-Shaped Stream and Eye Diagram (Tasks 23 & 27)



- **Impulse train:** 400 BPSK symbols, 32 samples/symbol.
- **After Tx RRC:** smooth band-limited waveform replaces the raw impulses.
- **After Rx RRC:** eye opens wide — traces converge at $t = T$ (red dashed line): the ISI-free sampling instant.



Bandwidth versus Roll-off — The Trade-off Curve



α	0	0.25	0.5	1
Occupied BW	$0.500 \cdot R_b$	$0.625 \cdot R_b$	$0.750 \cdot R_b$	$1.000 \cdot R_b$
Excess BW	0 %	25 %	50 %	100 %
Tail decay	$1/t$	$1/t^2$	$1/t^2$	$1/t^3$
Timing-jitter tolerance	very poor	poor	good	excellent

- **Rule:** $BW = (1+\alpha) \cdot R_b / 2$ — verified by the -6 dB edge of every simulated spectrum.
- **Engineering choice:** $\alpha = 0.35$ is common in practice (e.g. satellite/3G links); $\alpha = 0.2$ – 0.25 where spectrum is precious.

Mandatory Validation — RC Response Sampled at Integer Symbol Intervals

The overall RC response was sampled at $t = kT$ ($k = -5 \dots +5$) and tabulated — the defining test of the Nyquist criterion.

$k = t/T$	-5	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
$h(kT), \alpha=0$	0.0	-0.0	0.0	-0.0	0.0	1.0	0.0	-0.0	0.0	-0.0	0.0
$h(kT), \alpha=0.25$	0.0	-0.0	0.0	-0.0	0.0	1.0	0.0	-0.0	0.0	-0.0	0.0
$h(kT), \alpha=0.5$	-0.0	0.0	0.0	-0.0	0.0	1.0	0.0	-0.0	0.0	0.0	-0.0
$h(kT), \alpha=1$	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
$(g*g)(kT), \alpha=0.5$	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	1.0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0

Observation and Interpretation — Expected vs Observed

Parameter variation	Expected physical effect	Observation & agreement with theory
$\alpha = 0$ (sinc)	Minimum BW $0.5 \cdot R_b$; slow $1/t$ tail; high timing-jitter sensitivity	Brick-wall spectrum confirmed; pulse = ideal sinc. Agrees. Tiny truncation ripple at $ t > 6T$ noted (finite span).
$\alpha = 0.25 / 0.5$	25–50 % excess BW; tail decay $\sim 1/t^2$; better jitter tolerance	Spectral edge moves to $0.625 \cdot R_b / 0.75 \cdot R_b$ exactly as $BW = (1+\alpha)R_b/2$ predicts. Eye remains fully open. Agrees.
$\alpha = 1$	100 % excess BW; fastest $1/t^3$ decay; maximum robustness	Spectrum spans to $f \cdot T = 1$; zero crossings preserved; widest eye margin. Agrees.
RRC cascade	$g * g$ should reproduce RC and its zero crossings	Cascade trace overlaps ideal RC (deviation $< 0.5\%$). Agrees — validates matched-filter receiver design.
Zero crossings	$h(kT) = 0$ for all $k \neq 0$, any α	Tabulated samples = unit impulse within 10^{-6} . Agrees — Nyquist criterion verified numerically.

Comparison of Pulse-Shaping Strategies

Pulse	Zero ISI?	Occupied BW	Tail decay	Practical?	Use case
Rectangular (NRZ)	Yes (if unfiltered)	Unlimited ($1/f^2$)	hard edge	Only on ideal wideband channels	Baseband test, chip-to-chip
Sinc ($\alpha = 0$)	Yes	$0.5 \cdot R_b$ (minimum)	$1/t$ — slow	No — non-causal, jitter-sensitive	Theoretical reference
RC, $\alpha = 0.25\text{--}0.5$	Yes	$0.625\text{--}0.75 \cdot R_b$	$1/t^2$	Yes, with span $\approx \pm 8T$	Bandwidth-efficient links
RC, $\alpha = 1$	Yes	$1.0 \cdot R_b$	$1/t^3$ — fast	Yes, shortest filter	Robust mobile links
RRC (Tx only)	No — until matched filter	$(1+\alpha)/2 \cdot R_b$	$1/t$	Half of the shaping	Transmitter side
RRC * RRC	Yes	$(1+\alpha)/2 \cdot R_b$	$1/t^2$	Standard implementation	Nearly all modern modems

- **Takeaway 1:** zero ISI and minimum bandwidth (2 symbols/s/Hz) are only jointly possible in the $\alpha \rightarrow 0$ limit — every real system buys robustness with excess bandwidth.
- **Takeaway 2:** the RRC/RRC split gives the same end-to-end response as one RC filter, while halving filter length at each end and enabling matched-filter noise performance.

Conclusion

- **Filters built from first principles:** raised-cosine and root-raised-cosine pulses were generated directly from their mathematical definitions, with removable singularities ($t = 0, \pm T/2\alpha, \pm T/4\alpha$) handled by closed-form limits.
- **Roll-off sweep completed ($\alpha = 0, 0.25, 0.5, 1$):** impulse and frequency responses plotted; occupied bandwidth followed $BW = (1+\alpha) \cdot R_b/2$ exactly, confirming the bandwidth–timing trade-off.
- **Nyquist criterion verified:** the RC response sampled at integer symbol intervals equals the unit impulse $h(kT) = \delta[k]$; zero crossings confirmed for every roll-off factor.
- **Matched-filter cascade validated:** transmitter RRC convolved with receiver RRC reproduces the RC pulse ($g * g \approx h$), restoring zero-ISI sampling and an open eye diagram at the receiver.
- **Overall lesson:** pulse shaping lets a band-limited channel carry symbols at 2 symbols/s/Hz without ISI; the roll-off factor tunes the exchange between spectral efficiency and tolerance to timing jitter — the core idea behind every modern digital modem.

Thank You